Optical Spectroscopy — Session 01

Document 2 of 3 — Horiba Jobin Yvon FluoroMax-3 Spectrofluorometer

UNR Shared Instrumentation Laboratory, Room 016 | Monday, April 13, 2026

Prep once, run three times — read document 1 first

This is document 2 of 3 for Monday's session. Document 1 (UV-2550 Protocol) is the MASTER document and contains all the sample preparation (Part A), the packing list (Part B), and the safety checklist.

By the time you are reading this document at the bench, you should have already: (a) prepped all six samples on Sunday evening, (b) completed the UV-2550 scans and written down A_measured for each sample, and (c) computed the dilution factor D = A_measured / 0.05 for each fluorophore.

Do not prep samples here. Do not pack here. Do not re-state safety. Just run the FluoroMax.

Session sequence — where you are in the three-instrument run

1. UV-2550 (doc 1) — 09:30 – 10:15. ✓ completed before you pick up this document.

2. FluoroMax-3 (THIS document) — 10:30 – 11:45. Turn on the xenon lamp NOW if you have not already; it needs 20 minutes of warmup before any keeper scan.

3. Lambda 750 (doc 3) — 12:00 – 13:00. Same samples, rescanned on the research-grade double-beam plus extended into the NIR.

FluoroMax-3 session goal

Collect qualitative excitation and emission spectra for the six fluorophores using the excitation wavelengths derived from the UV-2550 session and the dilutions computed from the UV-Vis A values.

Establish a reproducible workflow for the FluoroMax-3 and export clean .csv files for archiving in the science/research repo.

Expected duration: 75 minutes at the instrument (plus 20 minutes xenon lamp warmup, which you start before coming back from the UV-2550).

Difficulty: Beginner fluorescence. The instrument does the hard part; your job is clean cuvettes and calm hands.

Safety reminder — FluoroMax-specific hazards

- The xenon arc lamp is a very bright broadband UV source. NEVER look directly into the sample chamber with the shutter open.
- PPE, flammables, and waste handling: see the master safety checklist in document 1, Part A header box.

Part A — Samples (see master document)

All six fluorophore samples (quinine, yellow HL, pink HL, curcumin, green tea extract, salicylate) plus both blank solvents (H2O, EtOH) were prepared on Sunday evening per document 1 (UV-2550 Protocol), Part A. Do not re-prep here.

At this point in the session you are also carrying six small 5 mL plastic bottles containing the DILUTED aliquots for fluorescence, prepared at the UV-2550 bench using the formula D = A_measured / 0.05. If you skipped the UV-2550 session for any reason, see the fallback dilutions in the sample parameter table in Part D below.

The FluoroMax scans use the DILUTED samples, not the original stocks. Keep both sets of bottles on the bench — the stocks for the Lambda 750 session immediately afterward.

Part B — FluoroMax-3 warmup and setup

You ideally turned the FluoroMax on as you left the UV-2550 room, so the xenon lamp has been warming up for ~15 minutes by the time you reach the bench. The xenon arc needs 15 – 30 minutes to stabilize before you should trust any intensity comparison. Use the remaining warmup window to clean cuvettes and configure the software.

Step C1 — Power on the FluoroMax-3

1. Locate the main power switch on the instrument (usually on the left side of the optical bench).
2. Turn it on. You should hear the lamp ignitor click and the cooling fan spin up.
3. Start the FluorEssence software (or DataMax, depending on which is installed) on the instrument PC. Let it connect to the instrument — the status bar will say "Ready" when handshaking is done.
4. Start a 20-minute timer. Do not run any scans you intend to keep during the first 15 minutes.

Step C2 — Cuvette cleaning ritual

Do this for BOTH fluorescence cuvettes before you touch any sample. The biggest rookie error in fluorescence is dirty cuvettes — even a fingerprint can double the background.

5. Hold the cuvette by its frosted top edge with the ceramic tweezers, never with bare fingers on the optical faces.
6. Rinse the inside 3 times with distilled water from the wash bottle. Dump waste into the waste bottle.
7. Rinse the inside 1 time with 95% ethanol. Dump waste.
8. Rinse the inside 1 more time with distilled water. Dump waste.
9. Gently invert on a folded Kimwipe to drain.
10. Polish each of the four optical faces with a fresh Kimwipe — one single wipe per face, no rubbing. Inspect under the room light for streaks, lint, or fingerprints. If you see any, repeat.
11. Dedicate one cuvette to "aqueous / standards" and the other to "dyes" (highlighters + turmeric). Label the frosted edge with a fine marker or a piece of lab tape if helpful.

Step C3 — Software configuration

In FluorEssence, set the following global defaults before running any sample. These apply to every scan unless overridden:

Parameter	Value	Why
Integration time	0.5 s per point	Good SNR without being slow
Excitation slit	2 nm	Moderate throughput, decent resolution
Emission slit	2 nm	Matches excitation slit
Step size	1 nm	Standard spectral resolution
Correction	Instrument correction ON (S1/R1)	Divides by lamp reference to remove xenon lamp spectral structure
Data format	Export as .csv per scan	Easy to load in Python later

If any sample shows a weak signal later, widen both slits to 3 or 4 nm and rescan. If any sample shows a saturated (flat-topped) signal, narrow slits to 1 nm or dilute the sample further.

Part C — Per-sample scan plan

For EVERY sample, do two scans in this order: (1) an emission scan at a fixed excitation wavelength, then (2) an excitation scan at a fixed emission wavelength. The two spectra together give you the characteristic fluorescence "fingerprint" for that molecule.

Run samples in this order (lowest-concentration to highest-concentration fluorophores first, to minimize cross-contamination risk):

12. Blank (water), then Sample 1 (Quinine)
13. Blank (water), then Sample 6 (Salicylate)
14. Blank (ethanol), then Sample 5 (Green tea extract)
15. Blank (ethanol), then Sample 4 (Curcumin)
16. Switch to the "dyes" cuvette. Blank (water), then Sample 2 (Yellow HL)
17. Blank (water), then Sample 3 (Pink HL)

The blanks are not just bureaucracy — they tell you whether the solvent is clean and give you a Raman/Rayleigh scatter reference for later background subtraction.

Scan parameters by sample

Two rows per sample: the first row is the emission scan (fix ex, sweep em), the second row is the excitation scan (fix em, sweep ex).

Sample	Scan type	Fixed λ	Scan range	Expected peak	Notes
1. Quinine	Emission	Ex = 350 nm	380 – 600 nm	~450 nm (blue)	Neat, no dilution
	Excitation	Em = 450 nm	250 – 430 nm	~335 nm	Watch for Raman water band ~395 nm
6. Salicylate	Emission	Ex = 300 nm	330 – 500 nm	~410 nm	Neat
	Excitation	Em = 410 nm	260 – 380 nm	~300 nm	Tablet binder may scatter slightly
5. Green tea	Emission	Ex = 430 nm	450 – 750 nm	~670 nm (red, chlorophyll)	Dilute 1 drop / 3 mL EtOH
	Excitation	Em = 670 nm	350 – 650 nm	~430 nm + ~660 nm	Dual Soret + Q band
4. Curcumin	Emission	Ex = 425 nm	450 – 700 nm	~540 nm (yellow-green)	Dilute 1 drop / 3 mL EtOH
	Excitation	Em = 540 nm	320 – 510 nm	~425 nm	Strong solvatochromism
2. Yellow HL	Emission	Ex = 488 nm	500 – 650 nm	~515 nm (green)	Dilute 1 drop / 3 mL H2O
	Excitation	Em = 515 nm	380 – 500 nm	~490 nm	If saturated, dilute 10× more

Sample	Scan type	Fixed λ	Scan range	Expected peak	Notes
3. Pink HL	Emission	Ex = 540 nm	555 – 700 nm	~580 nm (orange)	Dilute 1 drop / 3 mL H2O
	Excitation	Em = 585 nm	420 – 565 nm	~554 nm	Watch for inner-filter skew

Part D — Standard procedure for every sample

Repeat this 12-step cycle for each of the six samples. The first full cycle will feel slow; by sample 3 you will be flying.

18. In the software, create a new project folder named: 20260413_FluoroMax3_session01
19. Fill the appropriate clean cuvette (the aqueous/standards cuvette or the dyes cuvette, whichever matches the sample's solvent) with ~3 mL of the matching blank solvent.
20. Open the FluoroMax sample chamber lid. Using the ceramic tweezers, place the cuvette in the holder with the two polished faces perpendicular to the light path. Close the lid.
21. Run the emission scan from the parameter table for this sample. Name the file: 20260413_S{n}_BLANK_EM_ex{XXX}.csv
22. Run the excitation scan. Name the file: 20260413_S{n}_BLANK_EX_em{XXX}.csv
23. Open the lid, remove the cuvette, and empty the blank into the waste bottle.
24. Rinse the cuvette once with ~1 mL of the sample itself (a "pre-rinse" to displace residual blank). Discard that rinse.
25. Fill the cuvette with ~3 mL of the actual sample (pre-diluted per the table if required).
26. Wipe the four optical faces with a fresh Kimwipe. Inspect. Place in the holder.
27. Run the emission scan. File name: 20260413_S{n}_{sampleName}_EM_ex{XXX}.csv
28. Run the excitation scan. File name: 20260413_S{n}_{sampleName}_EX_em{XXX}.csv
29. Between-sample cleaning (see Part F). Move on to the next sample.

After each pair of scans, glance at the emission peak position and intensity. Spectra that look flat-topped, jagged, or show a strong rising tail at long wavelengths are usually too concentrated — dilute 10× with a fresh clean cuvette fill and rescan.

Part E — Between-sample cleaning

This is a 30-second ritual, not an exam. Do it every single time.

30. Dump the cuvette into the waste bottle.
31. Rinse 3× with distilled water. Dump each rinse.
32. Rinse 1× with 95% ethanol. Dump.
33. Rinse 1× with distilled water. Dump.
34. Drain on a Kimwipe, polish the four optical faces with a fresh Kimwipe.
35. Hold it up to the room light. If you can see a streak or a droplet, rewipe.

Extra step when switching from a strong dye (highlighters, turmeric) back to a weak fluorophore: add one ethanol rinse followed by one water rinse before the standard cycle. Residual rhodamine at the ppb level will still show up in a quinine emission scan.

Part F — Shutdown and handoff to Lambda 750

36. In FluorEssence, verify every .csv file you intended to save is actually on disk. Open at least two at random to confirm they are not empty.
37. Copy the entire 20260413_FluoroMax3_session01 folder to the USB stick.
38. Close FluorEssence.

39. Turn the FluoroMax-3 off at the main power switch. The cooling fan may continue running for a minute; that is normal — do not unplug.
40. Wipe down the benchtop with a fresh Kimwipe.
41. Do NOT pack out yet — you still have the Lambda 750 session ahead. Keep the six sample stocks, both blanks, the cuvettes, the wash bottles, and the waste bottle together and walk to the Lambda 750 bench.
42. The Lambda 750 session uses the original undiluted stocks (document 3, Part C), not the diluted FluoroMax aliquots. You can dispose of the diluted 5 mL bottles into the waste bottle now.
43. Log your visit per SIL room-access rules (time in / time out, instrument used).

Part G — Data management and archiving

Once home, commit the data to the research repository for reproducibility.

44. Copy the session folder into: GITHUB/science/research/20260413 FluoroMax-3/raw/
45. Create an index.md in that folder with: session date, instrument, operators (James + Vivian), software version, slit widths, list of samples, list of files, and any notes-from-the-bench.
46. Load one .csv into Python (pandas + matplotlib) to confirm it parses cleanly. Save one reference plot (quinine emission) as a PNG next to the index for the repo homepage.
47. Commit with message: "Add FluoroMax-3 first session — 6 fluorophores, qualitative scans".

Part H — Troubleshooting cheat sheet

Symptom	What to try
Flat baseline, no peak	Check lamp on; check shutter open; check cuvette is oriented correctly; check you are exciting at the right wavelength; widen slits to 3 nm
Huge peak at ex + ~50 nm	That is Raman scatter of water (ex + ~3400 cm^{-1}). Normal; it marks your excitation wavelength on the spectrum. Not a sample signal.
Flat-topped / clipped peak	Detector is saturated. Dilute sample 10× or narrow slits to 1 nm.
Peak shifts and gets weaker as concentration increases	Inner-filter effect. You are re-absorbing your own fluorescence. Dilute.
Noisy, jagged spectrum	Integration time too short. Increase to 1 s per point.
Peak at the Rayleigh line (= excitation wavelength) only	Scatter, not fluorescence. Sample may be cloudy (filter it) or the sample is truly non-fluorescent.
Quinine scan weak or missing	Tonic water was not degassed; bubbles scattered the signal. Decant fresh and wait.
Yellow HL peak in the wrong place	Different highlighter brands use different dye mixes. Scan a wider emission range (450–700 nm) to find the actual peak.
Curcumin spectrum looks different from the literature	Curcumin is strongly solvatochromic — the peak shifts by tens of nm depending on solvent polarity. Your ethanol peak near 540 nm is correct.
Software crashes	Note which scan crashed it. Reopen, reload last project, rerun. If repeat, try smaller step size or different export format.

Part I — Post-session reflection (for Vivian's notebook)

Questions to discuss while walking back to the car:

- Which sample had the largest Stokes shift (difference between excitation peak and emission peak)? What does that tell you about energy loss between absorption and emission?
- Quinine and salicylate both absorb near the UV (~300 nm) but emit at very different wavelengths. Why might that be?
- Why does curcumin in ethanol look different from curcumin in water (you can try water at home later)?
- Green tea has TWO excitation peaks for chlorophyll (~430 nm and ~660 nm). What does the second peak tell you about the electronic structure?
- Which of the six samples would be the best candidate for a quantitative follow-up experiment (concentration vs. emission intensity)? Which would be the worst, and why?

These are the real entry points into research-level thinking. Pick one that Vivian finds most interesting and plan a follow-up session around it.

— End of FluoroMax-3 protocol — continue to Lambda 750 (document 3) —

Horiba FluoroMax-3 Spectrofluorometer

Excitation-Emission Matrix (EEM) Contour Plot — Quick Guide

Software: FluorEssence (built on Origin) | **Alternate:** DataMax v2.20

What Is an EEM?

An Excitation-Emission Matrix (EEM) is a 2D fluorescence fingerprint of a sample. The instrument collects a series of emission scans at incrementally different excitation wavelengths, then stitches them into a matrix. The result is displayed as a contour plot: excitation wavelength on one axis, emission wavelength on the other, and fluorescence intensity as color. Each fluorescent compound appears as a distinct "island" on the map.

This is different from the individual 1D scans in Session 01 (which fix one wavelength and sweep the other). An EEM captures everything at once.

Prerequisites

1. **Lamp warmup.** Turn on the xenon lamp at least 20 minutes before scanning. The arc needs time to stabilize for consistent intensities across the full EEM.

2. **Clean cuvette.** Use a 4-clear-side quartz fluorescence cuvette. Follow the cleaning ritual: rinse 3× with DI water, 1× ethanol, 1× DI water, drain, polish all four faces with a fresh Kimwipe.

3. **Dilute sample.** Absorbance at the excitation wavelength should be < 0.1 to avoid inner-filter effects. If your sample is strongly colored, dilute further.

Setting Up the 3D / EEM Scan

4. **Open FluorEssence.** Launch the software on the instrument PC. Confirm the status bar says "Ready" (instrument is connected).

5. **Select 3D Scan mode.** From the experiment type menu, choose "3D Scan" (also labeled "EEM" in some versions). This is NOT the same as a regular emission or excitation scan.

6. **Set the parameters.** Configure the scan ranges and acquisition settings per the table below.

Recommended EEM Parameters

Parameter	Value	Notes
Excitation range	250 – 500 nm	Covers UV through visible excitation
Excitation step size	5 – 10 nm	5 nm for detail; 10 nm for a quick survey
Emission range	270 – 700 nm	Start 20 nm above lowest excitation to clear Rayleigh line
Emission step size	2 – 5 nm	Finer than excitation for better peak shape
Excitation slit	5 nm	Wider than 1D scans to keep scan time reasonable
Emission slit	5 nm	Match excitation slit
Integration time	0.1 – 0.5 s	0.1 s for bright samples; 0.5 s for dilute ones
Correction	S1/R1 (ON)	Removes xenon lamp spectral structure

Estimated scan time: ~20–40 minutes with 5 nm excitation steps. Coarser steps (10 nm) cut time roughly in half.

Running the Scan

7. Load the cuvette. Open the sample chamber, place the cuvette with two polished faces in the excitation beam path (perpendicular to the light). Close the lid.

8. Run a blank first. Fill the cuvette with pure solvent and run a quick EEM to see the Rayleigh and Raman scatter baseline. Save this — you can subtract it from sample data later.

9. Load sample and start. Rinse the cuvette, fill with your sample, and press Start. FluorEssence will show a live 3D progress display as it collects each excitation slice.

10. Wait for completion. Do not open the sample chamber during the scan. The software will notify you when done.

Generating the Contour Plot

11. View the result. FluorEssence (built on Origin) automatically displays the completed EEM. Use the menu to switch between 3D surface view and 2D contour map.

12. Adjust the color scale. Right-click the plot to adjust the color palette and intensity range. Narrowing the range can reveal weak peaks hidden by a dominant feature.

13. Slice the data. Use the data-slicing tools to extract individual emission or excitation spectra from any point on the contour map — useful for identifying specific peaks.

14. Export. Save as .csv for data analysis or export the contour image as PNG/PDF for reports.

Reading the Contour Plot

Your EEM will show several features:

A. Rayleigh scatter line. A bright diagonal where emission = excitation. This is scattered light, not fluorescence. It appears on every EEM.

B. Raman scatter line. A second diagonal offset by ~3400 cm^{-1} from the Rayleigh line (for water). Also not fluorescence.

C. Fluorescence peaks. Oval or round "islands" that appear off the diagonals. Each island is a fluorescent species in your sample. The position gives you excitation and emission maxima simultaneously.

TIP: For quinine (a great first test sample): you'll see one clean island centered around Ex ~350 nm / Em ~450 nm, well separated from the scatter lines.

Troubleshooting

CAUTION: If the entire plot is dominated by scatter lines with no visible fluorescence islands, your sample may not be fluorescent, or the concentration is too low. Try widening the slits to 8–10 nm or increasing integration time.

CAUTION: If peaks appear saturated (flat-topped islands), dilute the sample 10× or narrow the slits to 2–3 nm.

TIP: Subtract the blank EEM from the sample EEM in Origin/FluorEssence to remove the Rayleigh and Raman scatter lines, leaving only the fluorescence peaks.

References

FluorEssence User Manual v3.5:

horiba.com/fileadmin/uploads/Scientific/Downloads/UserArea/Fluorescence/Manuals/FluorEssence_V3_5_Manual.pdf

What Is an EEM? (Horiba explainer):

horiba.com/int/scientific/technologies/fluorescence-spectroscopy/what-is-an-excitation-emission-matrix-eem/

FluoroMax-3 Operation Manual with FluorEssence:

horiba.com/fileadmin/uploads/Scientific/Downloads/UserArea/Fluorescence/Legacy/FluoroMax-3___-P_Manual_with_FluorEssence-opt.pdf